

Fine-Grained Species Recognition on iNaturalist-2021: Traditional Baselines, Deep Learning, Interpretability, and Robustness Analysis

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Abstract

Accurate photographic species recognition is vital for biodiversity monitoring, yet fine-grained visual similarities pose steep challenges under limited data. We benchmark vision pipelines on a 1,000-species subset of iNaturalist-2021 (40 training images/class), comparing traditional handcrafted baselines (Random Forests with Colour Histograms/HOG, and Linear SVMs with BoVW-RootSIFT) against modern deep CNNs (EfficientNetV2 and ConvNeXt V2). Deep models achieve dramatic gains (ConvNeXt V2 reaching 81.00% Top-1 accuracy; EfficientNetV2 79.20%) over handcrafted baselines ($\leq 5.14\%$), establishing the necessity of transfer learning. Finally, post-hoc Grad-CAM analyses confirm models focus on authentic anatomical traits rather than background artifacts, while corruption benchmarks reveal notable vulnerability to motion blur despite strong resilience to JPEG compression.

1. Introduction

Automated species recognition from citizen-science imagery is vital for biodiversity monitoring and conservation [22]. As a canonical *fine-grained visual categorization* (FGVC) problem, it exhibits subtle inter-class morphological differences alongside severe intra-class variations in lighting, pose, and background clutter, compounded by low per-class training data.

We benchmark this challenge on a 1,000-species subset of iNaturalist-2021 mini [22] (40 train, 10 val, 10 test per species), systematically comparing two paradigms: (1) *traditional pipelines* using handcrafted features (Random Forests with color histograms/HOG, and BoVW-

RootSIFT with linear SVMs), and (2) *deep CNNs* using EfficientNetV2-S [20] and ConvNeXt V2-Tiny [23]. Across both architectures, we evaluate transfer learning vs. training from scratch alongside systematic augmentation and regularization ladders.

Beyond predictive accuracy, we investigate model reliability via two post-hoc analyses: (i) **Grad-CAM** [15] interpretability to verify anatomical focus across correct, incorrect, and confusable species pairs, and (ii) **ImageNet-C** corruption benchmarks [8] evaluating stability against noise, blur, and compression.

2. Literature Review

We review foundational and contemporary literature across feature representations, deep architectures, training strategies, and model evaluation for fine-grained visual categorization.

2.1. Handcrafted Baselines and Deep CNN Architectures

Early fine-grained visual categorization (FGVC) relied on handcrafted descriptors and classical classifiers. **Random Forests** combined global color histograms and texture descriptors (*e.g.*, HOG) to leverage color attributes alongside edge boundaries [6, 16, 18]. Concurrently, the **Bag-of-Visual-Words (BoVW)** framework encoded scale-invariant local keypoints (*e.g.*, SIFT [14]). **RootSIFT** [1] improved SIFT by L_1 -square-root normalization to approximate Hellinger kernel distances under Euclidean metrics. Codebooks were commonly enhanced via Spatial Pyramid Matching (SPM) [10], TF-IDF weighting, and soft assignment before linear **Support Vector Machine (SVM)** classification [3, 4].

Modern CNNs instead learn multi-scale hierarchies end-to-end. **EfficientNetV2** [20] uses compound scaling [19] with Fused-MBConv and squeeze-and-excitation blocks for parameter efficiency on species benchmarks [2]. More recently, **ConvNeXt V2** [23] modernized pure CNNs with large 7×7 depthwise kernels [11] and Global Response Normalization (GRN), optimizing inter-channel competition to capture fine-grained morphological details.

2.2. Transfer Learning, Augmentation, and Regularization

Under per-class data scarcity (40 images/species), **transfer learning** from ImageNet provides an indispensable prior, yielding faster convergence and superior generalization over training from scratch [27]. Both CNNs benefit from **AdamW** [13] optimization with linear warmup [7] and cosine learning rate schedules [12].

To mitigate overfitting, data augmentation is crucial [17]. Standard spatial/photometric transforms (random cropping, flipping, color jitter) provide basic invariances. Advanced regularization further boosts generalization: **Stochastic Depth** drops residual branches [9], **CoarseDropout** occludes rectangular patches [5], and sample-mixing methods (**MixUp** [25], **CutMix** [24]) construct convex linear combinations of inputs and targets to smooth decision boundaries.

2.3. Visual Explainability and Model Robustness

Deploying models in ecological settings requires explainability and robustness. **Grad-CAM** [15] computes gradient-weighted feature activations to verify whether predictions ground in authentic anatomy or background artifacts [26]. Concurrently, real-world deployment faces domain shifts from camera noise, blur, and compression. The **ImageNet-C** framework [8] evaluates test-time corruption resilience across severity levels without retraining.

3. Methodology

In this section, we detail our technical methodology for model evaluation, interpretability, and robustness assessment.

3.1. Comprehensive Method Development

3.1.1. Colour Histograms + HOG into Random Forest

The first model we developed was a traditional Random Forest classifier with colour histograms and HOG features extracted to create unique feature vectors for each image. The random forest classifier is a standard baseline that has been used in previous papers to achieve accurate, generalisable results for image classification [18]. Therefore, it was chosen as one of the main traditional classifiers to compare our more advanced models against.

Colour histograms represent the global distribution of pixel colours in images, and provide a very intuitive way

to understand the colour space of any image. For classifying fine-grained colourful visuals in mini-iNat2021, these colour histograms become vital providers of key discriminative information that allows the Random Forest classifier to differentiate different species classes via colour composition. For this task, HSV and L*a*b colour spaces were evaluated for the colour histograms as their channels are less correlated than RGB. L*a*b was tested against HSV to determine whether a more perceptually uniform colour space would yield a more discriminative feature space for colour representation across species.

HOG features, in particular, capture shape and general appearance of the image through edge and contour information [16]. Different species present different shapes and sizes, and so understanding these can help the classifier better distinguish between different species. Different cell pixel inputs were tested for the building of the HOGs primarily to examine whether smaller cells would yield finer-grained gradient information and therefore a more discriminative feature representation, at the cost of a higher-dimensional feature vector.

Feeding both the colour histograms and the histograms of oriented gradients into the random forest classifier, therefore, creates an excellent, generalisable baseline with accurate colour discrimination and shape/appearance discrimination to compare against deep learning methods.

3.1.2. BoVW–RootSIFT with Linear SVM

Our second traditional approach represents each image with a Bag-of-Visual-Words (BoVW) histogram [4] built from RootSIFT descriptors and classifies it with a linear Support Vector Machine (SVM). Unlike the global colour and gradient statistics of the Random Forest, this is a second hand-crafted representation based on local structure, giving a further point of comparison against the learned deep features.

We use SIFT [14] because its local descriptors are robust to scale and orientation changes common in citizen-science photographs, and RootSIFT [1] as a near cost-free refinement whose normalisation makes Euclidean comparison approximate the Hellinger kernel. We pair the resulting histograms with a linear SVM [3] because they are sparse and high-dimensional, a setting where linear classifiers perform well and scale to 40,000 training images across 1,000 classes. Keeping the classifier simple focuses the study on the quality of the representation.

Each ablation targets a known weakness of the plain histogram. Vocabulary size governs a granularity trade-off between merging distinct local structures and producing sparse, unstable histograms. Soft assignment addresses the quantisation error incurred by assigning each descriptor to a single word [21], a concern when species differ only in subtle detail. TF-IDF weighting tests whether words shared by most images are less discriminative here, and Spatial Pyramid Matching [10] restores the spatial layout the plain his-

togram discards. We vary one choice at a time, combine the beneficial ones, and finally tune the SVM regularisation parameter C . All choices are made on validation macro-F1, which weights every species equally and exposes weak per-class behaviour that overall accuracy can hide. Implementation details appear in Sec. 4.2.2.

3.1.3. EfficientNet V2

Our first deep-learning model is **EfficientNetV2-S** [20]. Following Sec. 2.1, its Fused-MBConv and MBConv design provides an efficiency-oriented deep baseline with learned visual features beyond the hand-crafted representations used by the traditional pipelines. The S variant keeps multiple controlled experiments practical while retaining the capacity needed for fine-grained classification across 1,000 species.

The subset provides only 40 training images per species, motivating the transfer-learning comparison in Sec. 2.2. We compare ImageNet-1K fine-tuning and training from scratch under matched no-augmentation and full-augmentation settings. Each regime uses the training schedule appropriate to fine-tuning or learning from random initialization, so the comparison evaluates the practical value of a generic visual prior when task-specific supervision is limited.

Following Sec. 2.2, the cumulative augmentation ablation adds Crop, Flip, and Color Jitter one at a time to identify each component’s contribution. Intermediate levels are evaluated in the pretrained regime, while the scratch regime uses the two endpoints to measure the full pipeline’s effect within the available compute budget. Within each initialization regime, optimization and regularization are held fixed across augmentation levels, allowing performance changes to be associated with augmentation. The implementation details are reported in Sec. 4.3.1.

3.1.4. ConvNeXt V2

We selected **ConvNeXt V2** [23] over a Transformer backbone for this study specifically because: (i) its convolutional inductive bias is well-suited to our computational resource constraints and the relatively small size of our subsampled dataset (40,000 training images), and (ii) its compatibility with fully-convolutional, gradient-based interpretability methods such as Grad-CAM (Sec. 3.2).

Fine-tuning and from-scratch training utilize standard optimization strategies inspired by Sec. 2.2; specific hyperparameter configurations and schedules are detailed in Sec. 12.

Motivated by Sec. 2.2, we ablate augmentation and regularization for ConvNeXt V2 in stages. Training begins from a shared Basic transform stack—random resized cropping, horizontal flipping, and colour jitter—after which the from-scratch runs successively enable DropPath, CoarseDropout, and CutMix/MixUp so that each addition can be assessed

in isolation. The pretrained comparison is limited to the None and Basic endpoints, whereas the fuller ladder is reserved for training from scratch. Aside from the DropPath rate and warmup schedule, which are themselves ablated, other optimization choices remain unchanged within each initialization setting. Details of the experimental setup appear in Sec. 4.3.2.

3.2. Advanced Method Development

We conduct two advanced studies on our best-performing ConvNeXt V2 model to evaluate visual interpretability and environmental robustness.

3.2.1. Grad-CAM Interpretability

To analyze spatial feature importance, we implement Grad-CAM [15] targeted at stage 4’s final depthwise conv layer (`dwconv`). For feature activation tensor $\mathbf{A} \in \mathbb{R}^{C \times H \times W}$ ($C = 768, H = 7, W = 7$) captured via PyTorch hooks, channel importance weights α_k^c and coarse heatmap $L_{\text{Grad-CAM}}^c$ are computed as:

$$\alpha_k^c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \frac{\partial Y^c}{\partial A_{i,j}^k} \quad (1)$$

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_{k=1}^C \alpha_k^c \mathbf{A}^k \right) \quad (2)$$

The single-channel heatmap is normalized to $[0, 1]$, bilinearly interpolated to input resolution, and blended with a `jet` colormap ($\alpha = 0.4$).

3.2.2. Robustness to Image Degradation

Following ImageNet-C principles [8], we evaluate test-set performance under 4 corruption types across 5 severity levels ($s \in \{0, 1, 2, 3, 4\}$, with $s = 0$ as baseline):

- **Additive Gaussian Noise:** $\mathbf{I}_{\text{noisy}} = \text{clip}(\mathbf{I} + \mathbf{N}, 0, 1)$ where $\mathbf{N} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.05s$.
- **Gaussian Blur:** 2D isotropic Gaussian filter with blur radius $r = 1.0s$ pixels.
- **Motion Blur:** Diagonal motion kernel $\mathbf{K} \in \mathbb{R}^{L \times L}$ ($L = 4s + 1, K_{i,j} = \frac{1}{L} \delta_{i,j}$) applied via `depthwise conv2d`.
- **JPEG Compression:** Encoded via PIL with quality $q(s) \in \{100, 50, 25, 10, 5\}$ for severity levels $s \in \{0, 1, 2, 3, 4\}$.

4. Experimental Results

We evaluate both traditional methods and deep learning approaches—under fine-tuning and training-from-scratch regimes—on a 1,000-species subsample of mini-iNat2021. Additionally, we conduct ablation studies on input features, hyperparameters, data augmentations, and regularization techniques.



Figure 1. **Visual demonstration of the 4 image degradation types at Severity Level 3.** From left to right: *Original Image* ($s = 0$), *Additive Gaussian Noise* ($\sigma = 0.15$), *Gaussian Blur* ($r = 3.0$), *Motion Blur* ($L = 13$), and *JPEG Compression* ($q = 10$).

4.1. Experimental setup

Dataset. We construct a 1,000-species subsample of mini-iNat2021 containing 50 images per class, totaling 50,000 images. For each class, 40 images are used for training and 10 images are used for validation (an 80%/20% train/validation split). An additional 10,000 unseen images are reserved exclusively for testing.

Metrics. We report balanced accuracy, Top-1 accuracy, Top-5 accuracy, and macro-averaged precision, recall, and F1 score.

Reproducibility. All experiments are conducted using the fixed random seed 20269517.

4.2. Traditional Methods

4.2.1. Random Forests

Feature Extraction Images are resized to dim (128, 128) before being passed into colour histogram and HOG feature extraction functions. A baseline feature space was first created (first rows of Tab. 1 and Tab. 2), followed by ablation studies that adjusted the colour space individually to L^*a^*b for the colour histograms while retaining the same HOG configuration as the base, and another study that adjusted the number of cell pixels for the HOG features while retaining the HSV colour space for the colour histograms. The configurations for both of these studies are shown in Tab. 1 and Tab. 2.

Table 1. Colour histogram configuration.

Colour Space	Bins per Channel	Value Range
HSV	$8 \times 8 \times 8$	H: [0, 180], S/V: [0, 256]
L^*a^*b	$8 \times 8 \times 8$	[0, 256] (all channels)

Table 2. HOG configuration.

Stage	Orientation	Cell Pixels	Block Cells
Base	9	8×8	2×2
Ablation	9	6×6	2×2

Modelling and Training Each of these feature spaces was separately fed into a random forest classifier with `n_estimators=150`, `max_depth=20`, `n_jobs=-1`, `warm_start=True`, `random_state=20269517`. The accuracy, precision, recall and F1 score for these are reported in Tab. 3.

From this, since mini-iNat2021 contains fine-grained images, putting more emphasis on the F1 score results in the baseline being the best model to move forward with. Using this baseline model, further training was carried out with `n_estimator` values of 200, 300, and 400 to see if more training improved accuracy. The results are shown in Tab. 4.

From this, it can be seen that additional estimators provide only marginal improvement, with the model at 400 estimators achieving the best validation F1 score. However, when evaluated on the test set, the model with 300 estimators achieved the best F1 score (recorded in Tab. 10), narrowly outperforming the 400-estimator model despite the latter’s slightly higher validation performance. Given that the difference between the 300- and 400-estimator models is minimal on both validation and test sets (accuracy and F1 differing by only 0.0004), this discrepancy is more likely due to the inherent randomness of Random Forest’s bootstrap sampling than to genuine overfitting, indicating that performance has effectively started to plateau from approximately 300 estimators.

Error Analysis Because the model trained with 300 estimators achieved the best macro-F1 score on the test data, it was selected for the remaining error analysis. First, the top 20 most confused species pairs were identified and visualised as a confusion matrix (spanning 30+ species classes in total; Fig. 3 in the supplementary material). This revealed that most confusion arises within insects and arthropods (*Arthropoda*), vertebrates (*Chordata*), sac fungi (*Ascomycota*), and vascular plants (*Tracheophyta*). A phylum-level confusion matrix (Fig. 4 in the supplementary material) clarified this further, showing substantial misclassification of arthropods (*Arthropoda*) as vascular plants (*Tracheophyta*), and of vertebrates (*Chordata*) as both arthropods (*Arthropoda*) and vascular plants (*Tracheophyta*). This may be partly explained by the fact that insects are frequently photographed while resting on or cam-

Table 3. Validation results for Random Forest feature configurations.

Model	Train Time (min)	(Bal.) Acc.	Top-1 Acc.	Top-5 Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
Baseline (HSV, HOG 8x8)	191.8	0.0319	0.0319	0.0999	0.0191	0.0319	0.0186
HOG diff (cells 6x6)	246.7	0.0281	0.0281	0.0880	0.0127	0.0281	0.0150
LAB colour space	185.1	0.0279	0.0279	0.0928	0.0157	0.0279	0.0158

Table 4. Validation results for baseline model trained with additional estimators.

Model	Add. Train Time (min)	(Bal.) Acc.	Top-1 Acc.	Top-5 Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
Baseline (200 estimators)	64.9	0.0340	0.0340	0.1014	0.0207	0.0340	0.0201
Baseline (300 estimators)	129.4	0.0358	0.0358	0.1087	0.0204	0.0358	0.0209
Baseline (400 estimators)	126.5	0.0395	0.0395	0.1131	0.0243	0.0395	0.0231

ouflaged among plants, causing the model to pick up on background vegetation rather than the insect’s own distinguishing features.

4.2.2. BoVW–RootSIFT with Linear SVM

Experimental setup. Following Sec. 4.1, development uses 40 training and 10 validation images per species, with 10 held-out test images per species for final evaluation. Vocabulary construction, representation selection, and C tuning use only the training and validation splits.

Features. Each image is resized so its longer side is 256 px and converted to greyscale. Sparse SIFT keypoints are detected, their descriptors converted to RootSIFT, and images with more than 500 descriptors keep the 500 strongest by keypoint response. Descriptors are extracted once and cached, so all variants share identical local features.

Vocabulary and encoding. The vocabulary is learned from training descriptors only, using mini-batch k -means (batch size 10,000, three initialisations, 100 iterations) on up to 500,000 sampled descriptors, with 1,024 words in the baseline. The baseline uses hard assignment, TF-IDF weighting, no spatial pyramid, and L2 normalisation. Soft assignment instead spreads each descriptor over its five nearest words with Gaussian weights whose bandwidth is the median training-descriptor-to-centre distance. The Spatial Pyramid Matching (SPM) variant concatenates one global histogram with four from a 2×2 grid (global weight 0.5, cells sharing the rest, then L2 normalisation), expanding the 1,024-word representation to 5,120 dimensions.

Protocol. Individual ablations use a linear SVM with $C = 1$ and sweep $K \in \{256, 512, 1024, 2048\}$, then soft assignment, TF-IDF removal, and SPM in isolation. Components that raise validation macro-F1 are combined at $K \in \{256, 1024\}$, and $C \in \{10^{-4}, \dots, 10^2\}$ is finally searched separately for the baseline and the selected representation, which are frozen before test evaluation.

Individual representation ablation. Tab. 5 reports

the individual experiments. Soft assignment is by far the strongest single change, raising macro-F1 from 2.62% to 3.96% and confirming that hard assignment introduces substantial quantisation error on this fine-grained task. SPM helps less (2.92%), so coarse spatial layout carries only a weak signal, and removing TF-IDF is essentially neutral (2.66%). Vocabulary size shows clear diminishing returns above 1,024 words, while shrinking it to 512 or 256 merges discriminative structures and hurts macro-F1.

Combined representation and C tuning. Combining the three beneficial components (Tab. 6) is strongest at 1,024 words (macro-F1 4.26%) and exceeds the best individual result (3.96%), so their gains are at least partially complementary. Tuning C selects 0.1 for the baseline (2.73%) and 1 for the improved representation, both frozen for test evaluation.

Held-out test results. On the held-out test set (Tab. 7), the improved representation raises macro-F1 from 2.65% to 4.26% and Top-1 from 3.63% to 5.14% over the tuned baseline, and its test macro-F1 matches validation (4.26%), so the selection transfers without over-fitting. This gain is costly: soft assignment and the 5,120-dimensional SPM features raise SVM training from 9.0 to 90.4 min and test encoding from 11.7 to 142.6 s. Even so, absolute performance stays low for 1,000-way fine-grained recognition, as visual-word histograms remain far less expressive than the learned deep features of Sec. 4.3.

Error analysis. The row-normalised confusion matrix over all 1,000 classes (Fig. 5 in the supplementary material) shows a faint diagonal against a diffuse background of errors, consistent with the low but above-chance accuracy: most images are misclassified and errors are spread widely rather than concentrated. Since the full matrix is dominated by this diffuse mass, we inspect the most-confused species subset (Fig. 5), which exposes clear structure. The strongest confusions fall between visually and taxonomically similar species: among cobweb spiders, *Steatoda grossa* is fre-

Table 5. Validation results for the individual BoVW–RootSIFT–SVM representation experiments. All configurations use a linear SVM with $C = 1$. SVM training time excludes SIFT extraction and visual-vocabulary construction.

Configuration	K	Encoding	TF-IDF	SPM	SVM Train Time (min)	(Bal.) Acc.	Top-1 Acc.	Top-5 Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
Baseline	1024	Hard	Yes	No	10.2	0.0294	0.0294	0.0723	0.0250	0.0294	0.0262
Vocabulary 256	256	Hard	Yes	No	6.1	0.0278	0.0278	0.0752	0.0175	0.0278	0.0196
Vocabulary 512	512	Hard	Yes	No	8.1	0.0286	0.0286	0.0724	0.0218	0.0286	0.0237
Vocabulary 2048	2048	Hard	Yes	No	11.0	0.0293	0.0293	0.0717	0.0272	0.0293	0.0272
Soft assignment	1024	Soft	Yes	No	31.6	0.0485	0.0485	0.1107	0.0368	0.0485	0.0396
No TF-IDF	1024	Hard	No	No	10.4	0.0297	0.0297	0.0718	0.0255	0.0297	0.0266
Spatial pyramid	1024	Hard	Yes	Yes	24.0	0.0325	0.0325	0.0764	0.0285	0.0325	0.0292

Table 6. Validation results for the combined BoVW configurations. Both configurations use soft assignment, no TF-IDF, Spatial Pyramid Matching, and $C = 1$.

Configuration	Vocabulary	Feature Dim.	Top-1 Acc.	Top-5 Acc.	F1 (macro)
Combined-256	256	1280	0.0470	0.1069	0.0359
Combined-1024	1024	5120	0.0509	0.1144	0.0426

quently predicted as *Eratigena duellica*; among seabirds and waterfowl with similar dark plumage, *Bucephala albeola*, *Melanitta deglandi*, and *Cerorhinca monocerata* are mutually confused; and pairs such as *Platanthera bifolia* with *Etlina elatior* follow the same pattern. This indicates that the BoVW–RootSIFT histogram captures coarse colour and shape cues well enough to group organisms of broadly similar appearance, but lacks the discriminative power to separate species differing only in subtle local detail, the limitation motivating the deep representations of Sec. 4.3. A few classes such as *Ardenna creatopus* retain a strong diagonal, showing the pipeline succeeds where inter-class differences are relatively large.

4.3. Deep Learning Methods

4.3.1. EfficientNet V2

Experimental setup. We train TorchVision EfficientNetV2-S on 384×384 inputs and replace its final layer with a 1,280-to-1,000 species classifier. All layers remain trainable. Pretrained runs use ImageNet-1K initialization and all four cumulative augmentation levels, while scratch runs use random initialization and the None and Full endpoints. The main EfficientNet V2 training settings for the pretrained and from-scratch regimes are summarized in Tab. 11 in the supplementary material.

Main results. Tab. 8 compares four cumulative augmentation levels for the pretrained model and two endpoints for the scratch model. Every class has exactly 10 test images, so balanced accuracy (macro recall) is identical to Top-1 accuracy and is not listed separately. C (pretrained Crop+Flip) records the highest test Top-1 and macro F1 at 79.20% and 79.08%, respectively.

Transfer learning vs. training from scratch. Without augmentation, pretraining raises Top-1 from 31.88% (E) to 77.63% (A), a 45.75-point gain. With full augmentation, it raises Top-1 from 48.52% (F) to 79.17% (D), a 30.65-point gain. Equivalently, removing transfer learning reduces Top-1 by 45.75 and 30.65 points in the two matched settings.

Effect of data augmentation. In the pretrained augmentation ablation, Crop raises Top-1 from 77.63% to 78.62% (+0.99 points), and Flip further raises it to 79.20% (+0.58). Adding Color Jitter changes Top-1 to 79.17% (−0.03), so C records the highest test Top-1. For scratch training, full augmentation raises Top-1 from 31.88% to 48.52% (+16.64) and macro F1 from 31.56% to 48.07% (+16.51). Ablating Crop or Flip reduces pretrained accuracy, whereas ablating Color Jitter increases it by 0.03 points. Removing the full augmentation pipeline lowers Top-1 for both pretrained (79.17% → 77.63%) and scratch (48.52% → 31.88%) training.

Training process dynamics. All six runs reach stable validation metrics within their epoch budgets and display sufficient convergence in the training curves, with some runs showing overfitting as training loss continues to fall after validation performance plateaus. For the representative Crop+Flip run, validation loss falls from 3.174 at epoch 5 to 1.848 at epoch 29 and ends at 1.850, while validation Top-1 varies by only 0.19 percentage points over epochs 25–30. The curves for all six runs are provided in Fig. 6 in the supplementary material.

Phylum-level error analysis. Aggregating C’s predictions by phylum gives 98.06% accuracy. The two highest row-normalized error cases are both for Echinodermata: 5.00% are predicted as Marchantiophyta and 5.00% as Mollusca, giving a total phylum error rate of 10.00% (2/20). Echinodermata therefore has the highest observed error percentage on this test set, based on 20 images. The count and row-normalized matrices are provided in Fig. 7 in the supplementary material.

Table 7. Held-out test results for the frozen baseline and selected improved BoVW–RootSIFT–SVM configurations.

Configuration	SVM Train Time (min)	Test Encoding Time (s)	(Bal.) Acc.	Top-1 Acc.	Top-5 Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
Baseline BoVW	9.0	11.69	0.0363	0.0363	0.0921	0.0243	0.0363	0.0265
Improved BoVW	90.4	142.59	0.0514	0.0514	0.1163	0.0397	0.0514	0.0426

Table 8. EfficientNetV2-S ablation results on the mini-iNat2021 test set. Train and test times are H:MM:SS.

ID	Init	Augmentation	Train Time	Test Time	Epochs	Top-1 Acc.	Top-5 Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
A	Pretrained	None	3:04:17	0:02:02	30	0.7763	0.9226	0.7909	0.7763	0.7742
B	Pretrained	Crop	3:13:57	0:00:53	30	0.7862	0.9290	0.8017	0.7862	0.7843
C	Pretrained	Crop+Flip	3:20:39	0:00:53	30	0.7920	0.9321	0.8074	0.7920	0.7908
D	Pretrained	Crop+Flip+Color Jitter	3:29:48	0:00:55	30	0.7917	0.9325	0.8076	0.7917	0.7903
E	Scratch	None	3:14:53	0:00:54	30	0.3188	0.5492	0.3474	0.3188	0.3156
F	Scratch	Crop+Flip+Color Jitter	5:31:07	0:00:54	50	0.4852	0.7055	0.4998	0.4852	0.4807

4.3.2. ConvNeXt V2

Experimental setup. We train ConvNeXt V2-Tiny on 224×224 inputs under two initialization regimes: (i) transfer learning with ImageNet-22K-pretrained weights, fully fine-tuned end-to-end; and (ii) training from scratch with randomly initialized weights. The augmentation and regularization strategies are applied incrementally on the from-scratch regime, adding one technique at a time. The main ConvNeXt V2 training settings for the pretrained and from-scratch regimes are summarized in Tab. 12 in the supplementary material. Representative training and validation loss curves for experiments B, C, H, and I are provided in Fig. 8 in the supplementary material. All experiments were run on a single NVIDIA T4 GPU via Google Colab.

Main results. Tab. 9 reports the ablation over initialization, cumulative augmentation level, DropPath, and warmup schedule across our 9 experiments (A–I). B (pretrained Basic) records the highest test Top-1 and macro F1 at 81.00% and 80.97%, respectively.

Transfer learning vs. training from scratch. Without augmentation, pretraining raises Top-1 from 12.87% (C) to 78.77% (A), a 65.90-point gain. With Basic augmentation, it raises Top-1 from 19.75% (D) to 81.00% (B), a 61.25-point gain.

Effect of data augmentation. On the pretrained regime, Basic augmentation alone (A $\rightarrow$ B) improves Top-1 from 78.77% to 81.00% (+2.23). On the from-scratch regime, the incremental ladder yields: Basic (C $\rightarrow$ D) +6.88 Top-1 (12.87% $\rightarrow$ 19.75%); DropPath at rate 0.1 (D $\rightarrow$ E) +1.85 (19.75% $\rightarrow$ 17.90%); CoarseDropout (E $\rightarrow$ F) +0.60 (17.90% $\rightarrow$ 18.50%); and batch-level CutMix/MixUp (F $\rightarrow$ G) +14.82 (18.50% $\rightarrow$ 33.32%).

Effect of warmup schedule. With the fully-augmented from-scratch configuration held fixed, the 10k-step warmup

(H) reaches 37.68% Top-1, versus 33.32% Top-1 for 1k (G) and 24.51% Top-1 for 30k (I).

Phylum-level error analysis. Aggregating B’s predictions by phylum gives 97.74% accuracy. Four major row-normalized error cases stand out: one for Ascomycota, in which 5.00% (5/100) are predicted as Tracheophyta, and three for Echinodermata, in which 5.00% are predicted as Basidiomycota, 5.00% as Cnidaria, and 5.00% as Mollusca. This gives Echinodermata a total phylum error rate of 15.00% (3/20). Echinodermata therefore has the highest observed error percentage on this 20-image test set. The count and row-normalized confusion matrices are provided in Fig. 9 of the supplementary material.

4.4. Overall Comparison

Tab. 10 summarizes the performance across all evaluated traditional and deep learning pipelines on the mini-iNat2021 test set. Hand-crafted feature baselines struggle to scale to 1,000-way fine-grained species categorization, with Bag-of-Visual-Words (BoVW-RootSIFT with Linear SVM) and Random Forest achieving Top-1 accuracies of only 5.14% and 3.64%, respectively. In contrast, modern deep learning architectures leverage learned hierarchical visual representations to achieve dramatic performance improvements: EfficientNetV2-S reaches 79.20% Top-1 accuracy (79.08% macro-F1), while ConvNeXt V2-Tiny achieves the highest overall performance at 81.00% Top-1 accuracy (80.97% macro-F1). This highlights the necessity of end-to-end deep learning features for complex fine-grained biological visual recognition.

4.5. Grad-CAM Qualitative Analysis

As an advanced analysis, we apply Grad-CAM to our best-performing model, ConvNeXt V2, to qualitatively probe what the network attends to. In this section, we present the qualitative results of our **Advanced Method Development** interpretability study using Grad-CAM. All the instances

Table 9. ConvNeXt V2 ablation results on the mini-iNat2021 test set. Train and test times are H:MM:SS.

ID	Init	Augmentation	DropPath Rate	Warmup Steps	Train Time	Test Time	Num. Epochs	Top-1 Acc.	Top-5 Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
A	Pretrained	None	0.0	1k	3:29:21	0:01:43	25	0.7877	0.9204	0.8030	0.7877	0.7869
B	Pretrained	Basic	0.0	1k	3:41:27	0:01:38	25	0.8100	0.9320	0.8244	0.8100	0.8097
C	Scratch	None	0.0	1k	1:59:19	0:01:40	14	0.1287	0.2984	0.1479	0.1287	0.1149
D	Scratch	Basic	0.0	1k	7:17:35	0:01:40	50	0.1975	0.3697	0.1874	0.1975	0.1860
E	Scratch	Basic	0.1	1k	3:17:15	0:01:39	22	0.1790	0.3753	0.2071	0.1790	0.1728
F	Scratch	Basic+CD	0.1	1k	3:43:46	0:01:39	25	0.1850	0.3730	0.2098	0.1850	0.1790
G	Scratch	Basic+CD+Mix	0.1	1k	8:33:00	0:01:39	58	0.3332	0.5492	0.3581	0.3332	0.3254
H	Scratch	Basic+CD+Mix	0.1	10k	18:09:21	0:01:40	112	0.3768	0.5886	0.4137	0.3768	0.3689
I	Scratch	Basic+CD+Mix	0.1	30k	6:51:06	0:01:40	47	0.2451	0.4489	0.2734	0.2451	0.2357

Table 10. Overall comparison of all evaluated methods on the mini-iNat2021 test set.

Method	Family	Train (min)	Test (min)	(Bal.) Acc.	Top-1 Acc.	Top-5 Acc.	Prec.	Recall	F1
BoVW–RootSIFT–Linear SVM	Traditional	90.40	2.38	0.0514	0.0514	0.1163	0.0397	0.0514	0.0426
Random Forests (300)	Traditional	386.1	0.9	0.0364	0.0364	0.1061	0.0239	0.0364	0.0226
EfficientNet V2	Deep Learning	200.65	0.88	0.7920	0.7920	0.9321	0.8074	0.7920	0.7908
ConvNeXt V2	Deep Learning	221.45	1.63	0.8100	0.8100	0.9320	0.8244	0.8100	0.8097

are randomly chosen, and composite visual strips containing the *Input Image*, *Normalized Heatmap*, and *Blended Overlay* are generated to inspect spatial attention.

4.5.1. Part 1: Correct Predictions (Semantic Feature Localization)

To verify whether the fine-tuned ConvNeXt V2 model relies on genuine biological morphology rather than background artifacts, we inspect activation heatmaps for randomly sampled correctly classified test images (Fig. 2).

As illustrated in Fig. 2, for *Pachodynerus erynnis* (Fig. 2a), the peak activation hotspot (red/yellow) is tightly concentrated along the red-and-black abdominal segments and thoracic thorax. Similarly, for *Scaevola taccada* (Fig. 2b), the network highlights the distinctive fan-shaped white petals and glossy leaf cluster. This confirms that the model has learned meaningful anatomical features rather than exploiting background foliage shortcuts.

4.5.2. Part 2: Failure Cases (Incorrect Predictions Diagnostic)

Understanding misclassifications is essential for identifying model vulnerabilities. Fig. 2 presents Grad-CAM visual explanations for failure cases where top-1 predictions diverged from ground-truth labels.

Our diagnostic analysis of Fig. 2 reveals two primary failure modes:

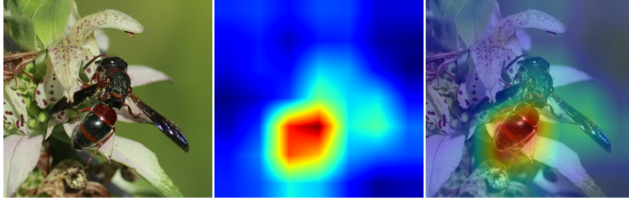
1. **Fine-Grained Intra-Genus Ambiguity:** In Fig. 2c, the ground-truth specimen is *Wahlenbergia marginata* (Southern Bluebell), but the model predicted its sister species *Wahlenbergia albomarginata*. The model correctly localized the flower genus, but extreme visual overlap between floral petals under outdoor glare caused confusion between species within the same genus.
2. **Habitat Forest-Floor Distraction:** In Fig. 2d, the target

Caloscypha fulgens (a small cup fungus) is surrounded by dense forest moss and leaf litter. Gradient energy misdirected onto background moss textures rather than the small fungal fruiting body, resulting in misclassification as lichen.

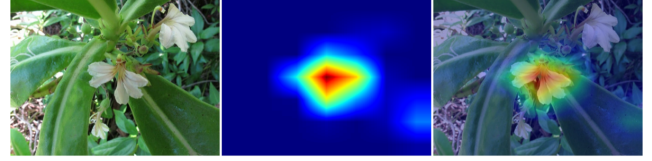
4.5.3. Part 3: Confusable Species Pairs (Fine-Grained Discrimination)

A key challenge in biodiversity monitoring is discriminating morphologically similar species from the same genus. We evaluate Grad-CAM across three confusable pairs selected via randomized image sampling across distinct biological classes: Spiders (*Steatoda*), Bumblebees (*Bombus*), and Mint Moths (*Pyrausta*). As illustrated in the supplementary material (Sec. 15, Fig. 10), ConvNeXt V2 correctly classifies all six species instances across the three confusable pairs:

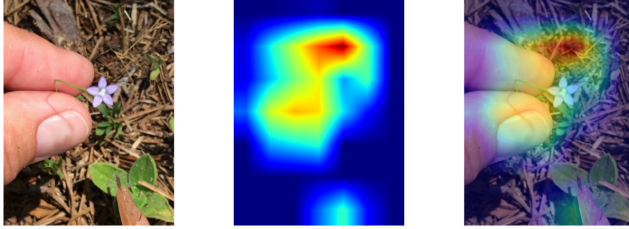
- **Genus *Steatoda*:** For *S. capensis* vs. *S. grossa*, the detection primarily targets the body and head, while extending to include the spider’s legs for both images, which indicate that the model can classify similar-looking but different species according to their morphological and anatomical features.
- **Genus *Bombus*:** For *B. huntii* vs. *B. mixtus*, the attention on the first image is drawn mainly on the head of the bee, while the second image targets not only on its head but also several parts of the body. Notably, due to the noise background in the first image, the attention was also drawn by the flowers surrounded, as we can tell from the heatmap.
- **Genus *Pyrausta*:** For *P. purpuralis* vs. *P. aurata*, activation maps isolate vibrant purple-gold wing patterns on *P. purpuralis* and central golden-yellow forewing spots on *P. aurata*, demonstrating fine-grained anatomical discrimination across distinct insect groups.



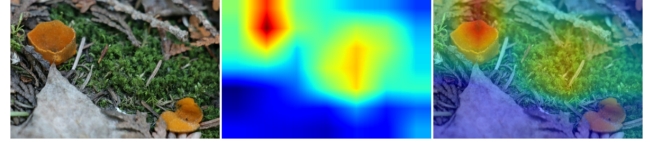
(a) Correct Sample 1: *Pachodynerus erynnis* (Wasp, Top-1 Prob: 0.8703). High activation on abdomen and thorax.



(b) Correct Sample 2: *Scaevola taccada* (Beach Cabbage, Top-1 Prob: 0.1133). Focus on petal fan structure.



(c) Failure Case 1: Ground-truth *Wahlenbergia marginata* misclassified as near-relative *W. albomarginata* (Prob: 0.0356).



(d) Failure Case 2: Ground-truth cup fungus *Caloscypha fulgens* misclassified as lichen *Peltigera membranacea* (Prob: 0.0310).

Figure 2. **Grad-CAM visual explanations for correct predictions and failure cases.** Top row (Figs. 2a and 2b): Correctly classified samples showing high activation (red/yellow) conforming tightly to morphological boundaries (abdomen, thorax, petals). Bottom row (Figs. 2c and 2d): Misclassified samples illustrating fine-grained intra-genus ambiguity and complex forest-floor background distraction.

4.6. Image Degradation Robustness Evaluation

We evaluate the fine-tuned ConvNeXt V2 model across all 1,000 fine-grained species classes under synthetic image quality corruptions across 5 severity levels ($s \in [0, 4]$) as specified in Sec. 3.2.2. Fig. 11 plots the resulting Top-1 Accuracy and Macro-averaged F1 Score performance curves.

On uncorrupted clean images ($s = 0$), the model achieves a baseline Top-1 Accuracy of **77.80%** and a Macro F1 score of **73.01%**. As demonstrated in Fig. 11, performance degrades monotonically across all 4 degradation types, revealing distinct levels of architectural sensitivity:

- **JPEG Compression (Highest Resilience):** ConvNeXt V2 exhibits exceptional tolerance to compression artifacts. At Level 1 ($q = 50$) and Level 2 ($q = 25$), Top-1 accuracy remains high at 76.00% and 73.00% respectively. Even under heavy compression at Level 3 ($q = 10$), the model retains 64.50% accuracy (57.95% Macro F1), before dropping to 43.10% at extreme compression ($q = 5$). This demonstrates that deep convolutional feature maps are robust to high-frequency 8×8 DCT block quantization noise.
- **Additive Gaussian Noise (Moderate Linear Degradation):** Adding Gaussian pixel noise leads to a steady, nearly linear decline in accuracy, moving from 73.80% at Level 1 ($\sigma = 0.05$) to 50.70% at Level 4 ($\sigma = 0.20$, Macro F1 of 43.43%). Because depthwise 7×7 convolutions aggregate local spatial context, the network filters out moderate pixel-level variance better than fine blur.
- **Gaussian Blur (Steep Defocus Decay):** Defocusing induces sharper degradation than noise. While mild blur

($r = 1.0$) retains 75.00% accuracy, performance drops to 63.00% at Level 2 ($r = 2.0$) and degrades heavily to 35.00% at Level 4 ($r = 4.0$, Macro F1 of 28.67%). Spatial smoothing erodes high-frequency edge gradients required for fine-grained morphological identification.

- **Motion Blur (Most Destructive Distortion):** Directional motion streak convolution causes the steepest performance collapse, with accuracy dropping from 70.00% at Level 1 ($L = 5$) down to 52.00% at Level 2 ($L = 9$) and reaching a low of **26.80%** at Level 4 ($L = 17$, Macro F1 of 21.25%). Because motion blur smears diagnostic biological structures (e.g., insect leg segments, wing veins, and floral stamen) along a diagonal trajectory, local depthwise spatial kernels fail to extract discriminative cues.

5. Discussion

5.1. Comparison with Traditional Methods

Comparing traditional baselines (Tabs. 3 and 7) highlights local vs. global feature limits. Random Forest using global HSV histograms and HOG achieves only 3.95% Top-1 (2.31% macro-F1) due to background and lighting sensitivity. Local RootSIFT keypoints with BoVW, soft assignment, and SPM reach 5.14% Top-1 (4.26% macro-F1, Tab. 7), but at heavy computational cost (SVM training time rises from 9.0 to 90.4 min, test encoding from 11.7 to 142.6 s). Handcrafted features thus scale poorly for 1,000-class fine-grained tasks.

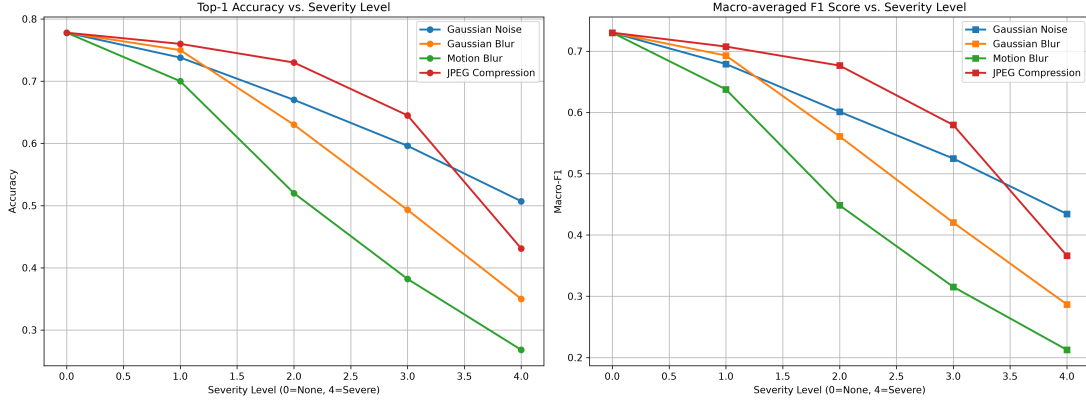


Figure 3. **Performance-versus-severity curves across 1,000 species under 4 image degradation types.** Left: Top-1 Accuracy vs. Severity Level (0 = Clean, 4 = Severe). Right: Macro-averaged F1 Score vs. Severity Level. Baseline uncorrupted images achieve 77.80% Top-1 Accuracy and 73.01% Macro F1.

5.2. Deep Learning Method Comparison

EfficientNetV2-S and ConvNeXt V2-Tiny (Tabs. 8 and 9) show that ImageNet pretraining is essential under low per-class data (40 images/species). Pretrained models perform best under light-to-moderate augmentation (79.20% Top-1 for EfficientNetV2 Crop+Flip; 81.00% for ConvNeXt V2 Basic). From-scratch models severely underfit without heavy regularization, improving from 12.87% (unaugmented) to 48.52%–51.78% with CutMix/MixUp and optimal 10k-step warmup, yet remaining > 29 percentage points below pretrained fine-tuning.

5.3. Overall Comparison

A cross-paradigm comparison (Tab. 10) reveals an overwhelming performance divide exceeding 75 percentage points between handcrafted methods and deep CNNs. Traditional pipelines (Random Forest at 3.64% Top-1 and BoVW RootSIFT-SVM at 5.14%) suffer from rigid feature spaces and orderless visual-word codebooks that lack the spatial hierarchies required to disambiguate 1,000 taxa in natural habitats. Under severe data scarcity (40 images/species), ImageNet pretraining proves indispensable: pretrained backbones achieve 79.20% (EfficientNetV2-S) and 81.00% (ConvNeXt V2-Tiny) Top-1 accuracy, substantially outpacing the strongest regularized from-scratch models (51.78%).

Between deep architectures, ConvNeXt V2-Tiny marginally outperforms EfficientNetV2-S (81.00% vs. 79.20% Top-1; 80.97% vs. 79.08% macro-F1). Its 7×7 depthwise convolutions and Global Response Normalization (GRN) enhance inter-channel feature diversity, preserving subtle localized morphological cues (*e.g.*, wing venation and petal geometry) more effectively. Moreover, the tight alignment between Top-1 accuracy and macro-F1 across both models confirms that performance gains

generalize uniformly across all 1,000 species.

5.4. Interpretability and Robustness Insights

Grad-CAM heatmaps (Sec. 4.5) confirm ConvNeXt V2 attends to anatomical morphology (*e.g.*, abdominal markings, petal fan structures, wing spots) rather than background context. Misclassifications stem from fine-grained intra-genus ambiguity or forest clutter (Fig. 2). Robustness evaluation (Sec. 4.6) shows high resilience to JPEG compression (64.50% Top-1 at Level 3, $q = 10$) due to convolutional pooling over block noise, but high vulnerability to motion blur (26.80% at Level 4) and defocus blur (35.00%) as spatial smoothing destroys sharp diagnostic edges.

5.5. Limitations and Future Directions

Key limitations include heavy reliance on ImageNet pretraining under data scarcity (40 images/class) and vulnerability to field photographic blurs. Future work includes blur-aware training augmentations and self-supervised domain adaptation.

6. Conclusion

We conducted a systematic empirical study of fine-grained 1,000-species classification on iNaturalist-2021 under data-constrained supervision (40 images/class). Our results demonstrate the necessity of ImageNet transfer learning, with modern CNNs (ConvNeXt V2 at 81.00% Top-1) vastly outperforming handcrafted baselines (5.14%). Grad-CAM visual interpretability confirms models focus on diagnostic anatomical traits rather than background context, while corruption benchmarks reveal pronounced vulnerabilities to motion and defocus blurs despite resilience to digital compression. These findings provide actionable insights for deploying robust ecological vision systems and highlight blur-resilient augmentation as a key future direction.

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**Fine-Grained Species Recognition on iNaturalist-2021: Traditional Baselines,
Deep Learning, Interpretability, and Robustness Analysis**

Supplementary Material

7. Colour Hists + HOG + Random Forest

The following matrices correspond to the classification performance of the Random Forest method.

The first matrix shows the species codes for the 20 most commonly confused species pairs by the model (30+ total species) Fig. 3.

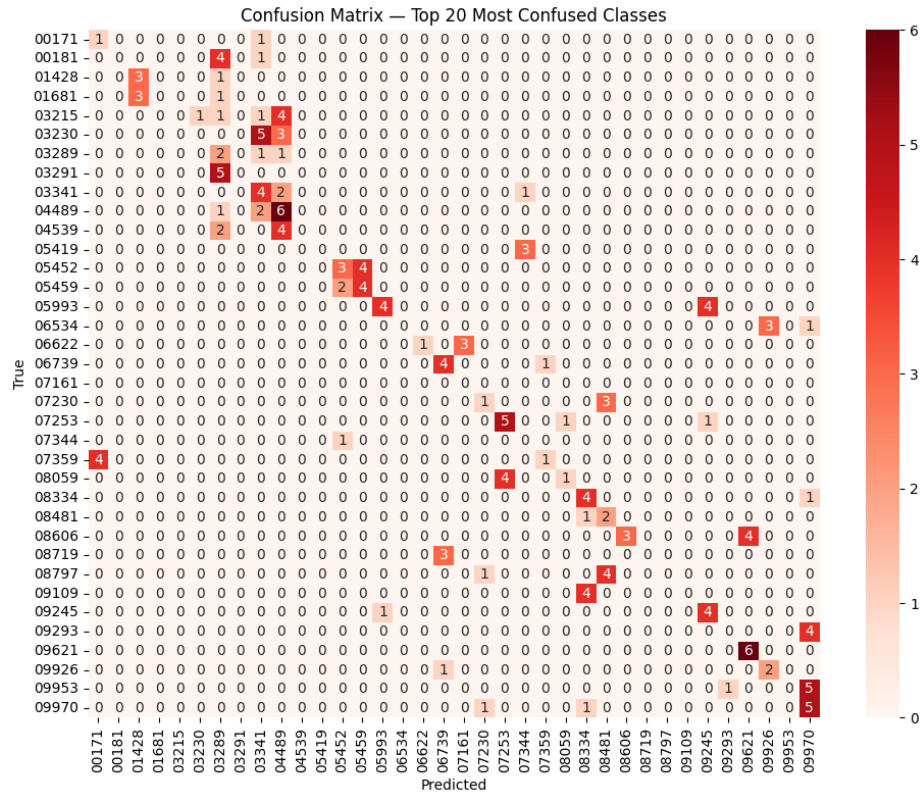


Figure 4. Confusion matrix of the top 20 most confused class pairs.

The second matrix shows the classification accuracy on a phylum level Fig. 4.

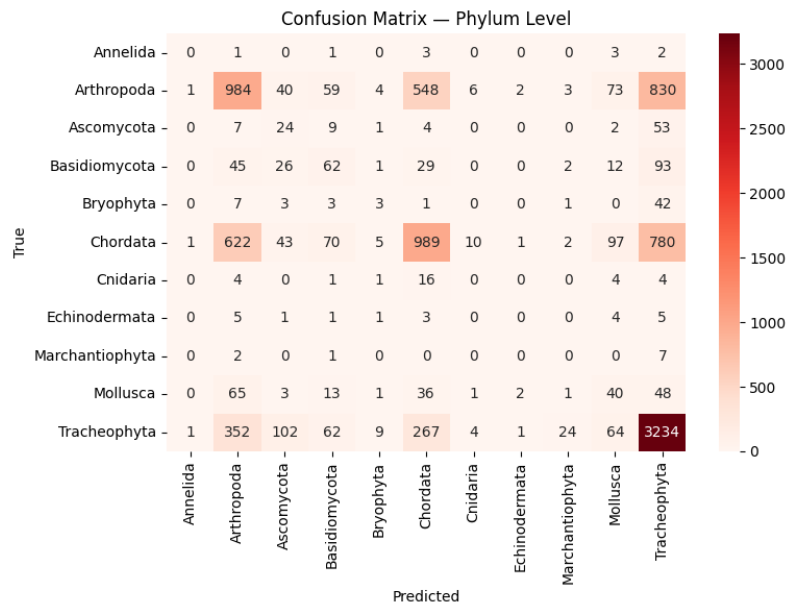


Figure 5. Confusion matrix at the phylum level.

8. BoVW–RootSIFT–SVM confusion matrices

The following confusion matrices show the classification performance of the improved BoVW–RootSIFT–SVM representation. The first matrix includes all 1,000 species, while the second focuses on the most-confused species.

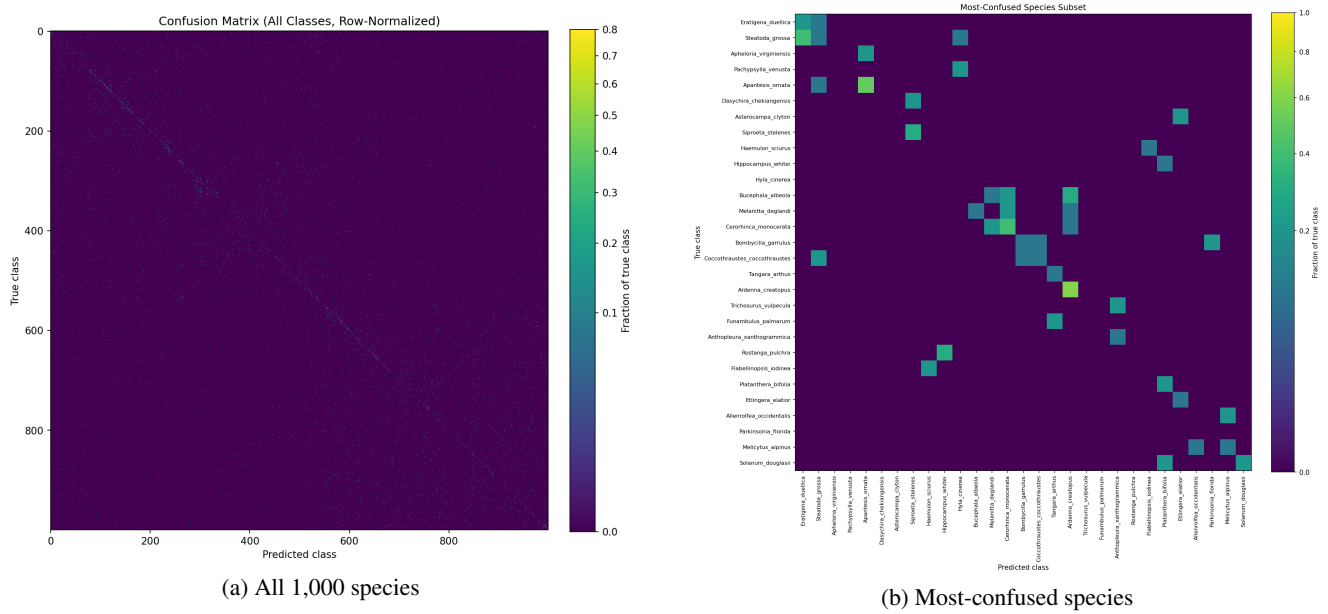


Figure 6. Confusion matrices for the improved BoVW–RootSIFT–SVM representation. (a) Row-normalised confusion matrix over all 1,000 species. The faint diagonal against a diffuse error background reflects the low but above-chance accuracy of a hand-crafted pipeline on a 1,000-way fine-grained classification task. (b) Confusion matrix restricted to the most-confused species. The strongest off-diagonal cells occur between visually and taxonomically similar species, including cobweb spiders *Eratigena/Steatoda* and dark-plumaged seabirds and waterfowl *Bucephala/Melanitta/Cerorhinca*.

9. EfficientNet V2 training hyperparameters

Tab. 11 lists the training hyperparameter settings for the EfficientNet V2 pretrained and from-scratch regimes described in Sec. 4.3.1.

Table 11. EfficientNet V2-S training hyperparameters for pretrained and from-scratch runs. Settings marked “ablated” are varied across the six experiments in Tab. 8. Rows marked (*aug.*) are data augmentation components.

Setting	Pretrained	From scratch
Optimizer	AdamW	AdamW
LR schedule	Cosine decay + linear warmup	Cosine decay + linear warmup
Initial learning rate	5×10^{-5}	1×10^{-3}
Weight decay	0.05	0.05
Stochastic depth	0.2 (fixed)	0.2 (fixed)
Classifier dropout	0.2	0.2
Augmentation level (ablated)	None / Crop / Crop+Flip / Full	None / Full
RandomResizedCrop (<i>aug.</i>)	scale = (0.7, 1.0)	scale = (0.7, 1.0)
HorizontalFlip (<i>aug.</i>)	$p = 0.5$	$p = 0.5$
ColorJitter (<i>aug.</i>)	b/c/s = 0.2, hue = 0.05, $p = 0.5$	b/c/s = 0.2, hue = 0.05, $p = 0.5$
Warmup steps	1,000 (fixed)	1,000 (fixed)
Early-stopping patience	5	5
Early-stopping threshold	0.0	0.0
Batch size	128 (effective)	128 (effective)
Label smoothing	0.1	0.1

10. EfficientNet V2 training curves

Fig. 6 shows training and validation loss for all six runs in Tab. 8.

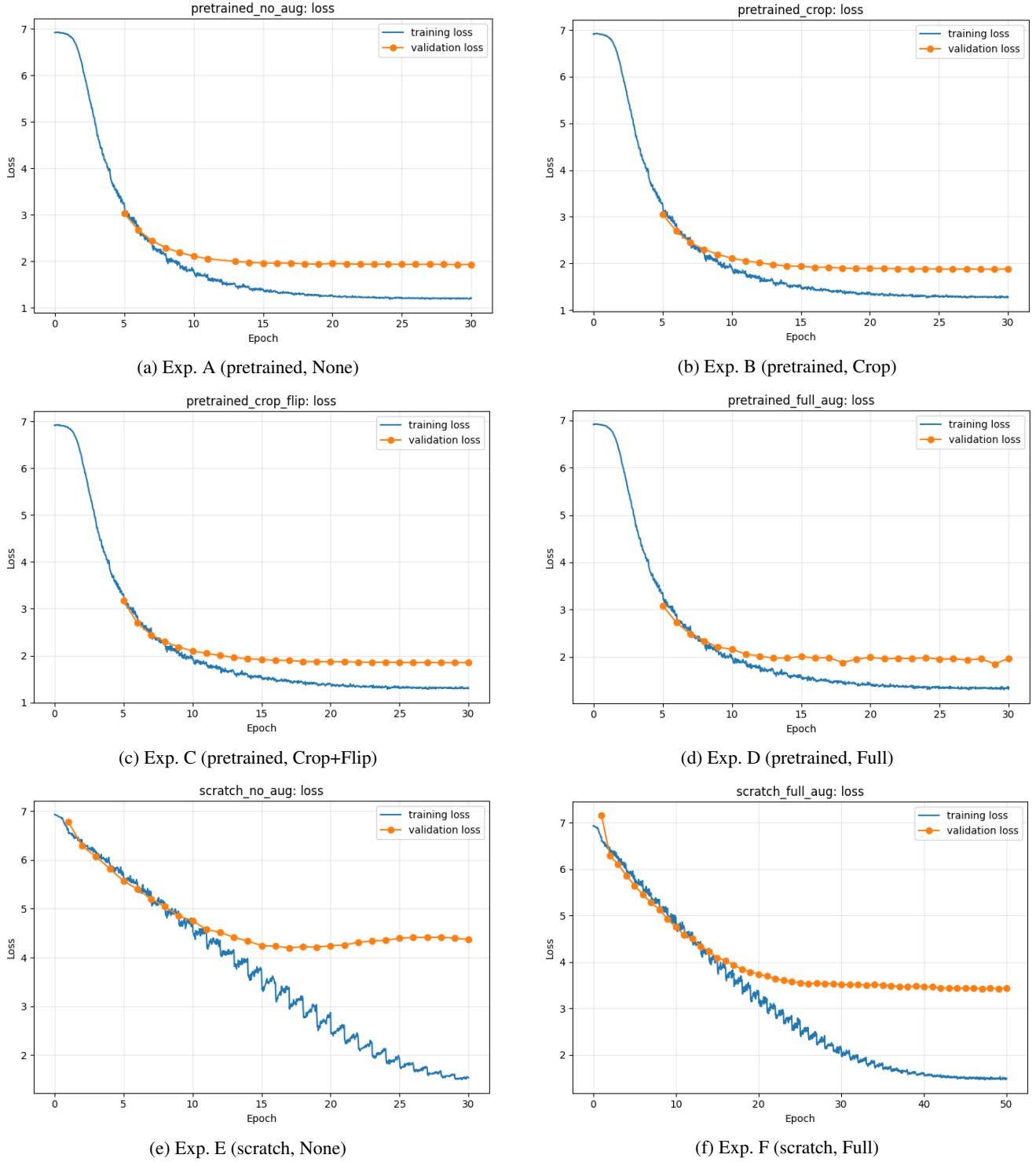


Figure 7. EfficientNet V2 training and validation loss curves for experiments A–F from Tab. 8.

11. EfficientNet V2 phylum-level confusion matrices

Fig. 7 shows phylum-aggregated predictions for experiment C from Tab. 8, corresponding to the analysis in Sec. 4.3.1.

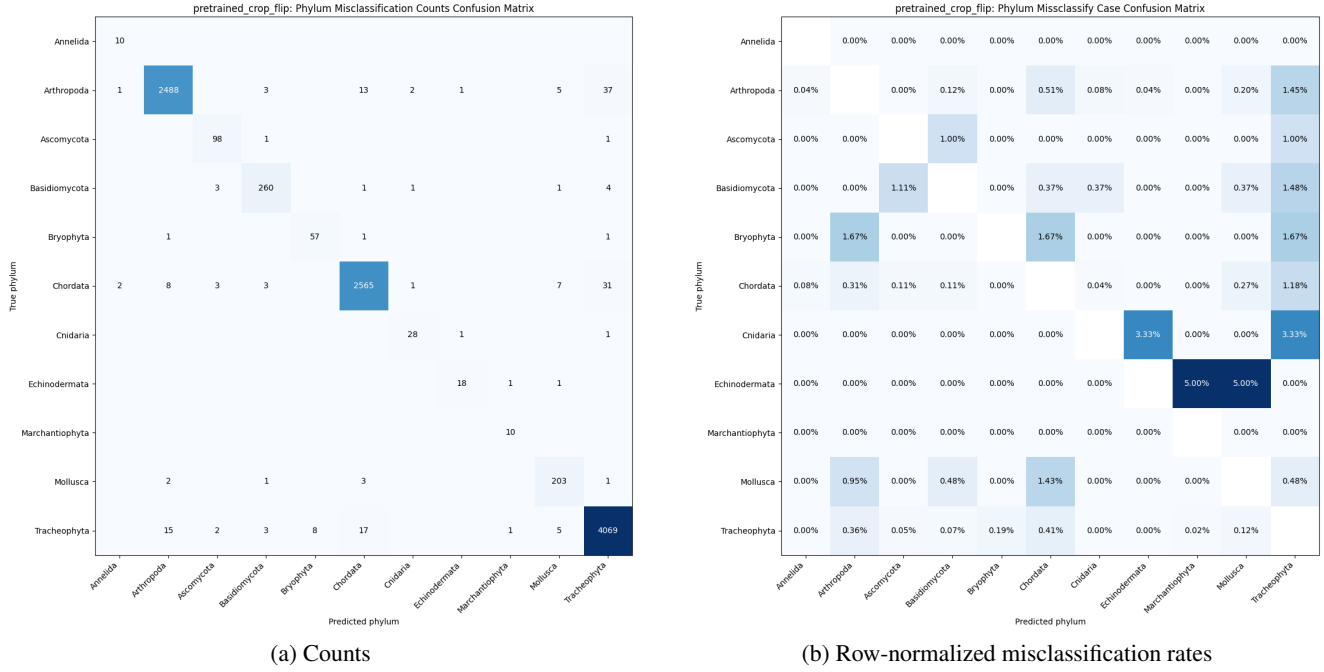


Figure 8. Phylum-level confusion matrices for EfficientNet V2 experiment C. (a) Absolute counts including correct predictions on the diagonal. (b) Row-normalized off-diagonal misclassification rates.

12. ConvNeXt V2 training hyperparameters

Tab. 12 lists the full training hyperparameter settings for the ConvNeXt V2 pretrained and from-scratch regimes described in Sec. 4.3.2.

Table 12. ConvNeXt V2 training hyperparameters for pretrained and from-scratch runs. Settings marked “ablated” are varied across the 9 experiments in Tab. 9. Rows marked (*aug.*) are data augmentation components; +*CD*/+*Mix* indicate the ablation tier introducing each.

Setting	Pretrained	From scratch
Optimizer	AdamW	AdamW
LR schedule	Cosine decay + linear warmup	Cosine decay + linear warmup
Initial learning rate	5×10^{-5}	1×10^{-3}
Weight decay	0.05	0.05
DropPath	0 (fixed)	0 / 0.1 (ablated)
Augmentation level (ablated)	None / Basic	None / Basic / Basic+CD / Basic+CD+Mix
RandomResizedCrop (<i>aug.</i>)	scale = (0.7, 1.0)	scale = (0.7, 1.0)
HorizontalFlip (<i>aug.</i>)	$p = 0.5$	$p = 0.5$
ColorJitter (<i>aug.</i>)	b/c/s = 0.2, hue = 0.05, $p = 0.5$	b/c/s = 0.2, hue = 0.05, $p = 0.5$
CoarseDropout (<i>aug.</i> , +CD)	–	holes = (3, 8), size = (0.05, 0.15), $p = 0.5$
CutMix/MixUp (<i>aug.</i> , +Mix)	–	batch-level
Warmup steps	1000 (fixed)	1000 / 10000 / 30000 (ablated)
Early-stopping patience	5	5
Early-stopping threshold	0.0	0.0
Batch size	128	128
Label smoothing	0.1	0.1

13. ConvNeXt V2 training curves

Fig. 8 shows training and validation loss versus epoch for a curated subset of the A–I ablation in Tab. 9: the best pretrained run (B), the scratch baseline without augmentation (C), the best from-scratch run (H), and the over-long warmup failure case (I).

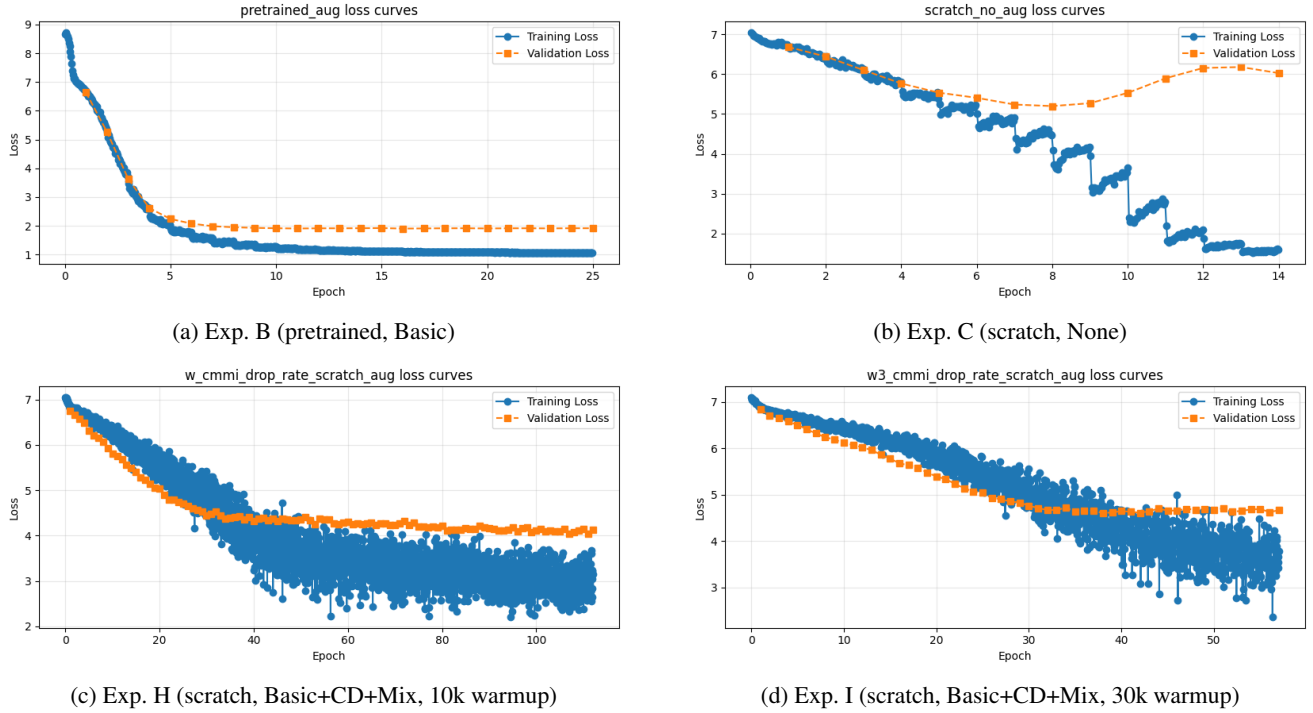


Figure 9. ConvNeXt V2 training (solid) and validation (dashed) loss curves for experiments B, C, H, and I from Tab. 9.

14. ConvNeXt V2 phylum-level confusion matrices

Fig. 9 shows phylum-aggregated predictions for experiment B from Tab. 9, corresponding to the analysis in Sec. 4.3.2.

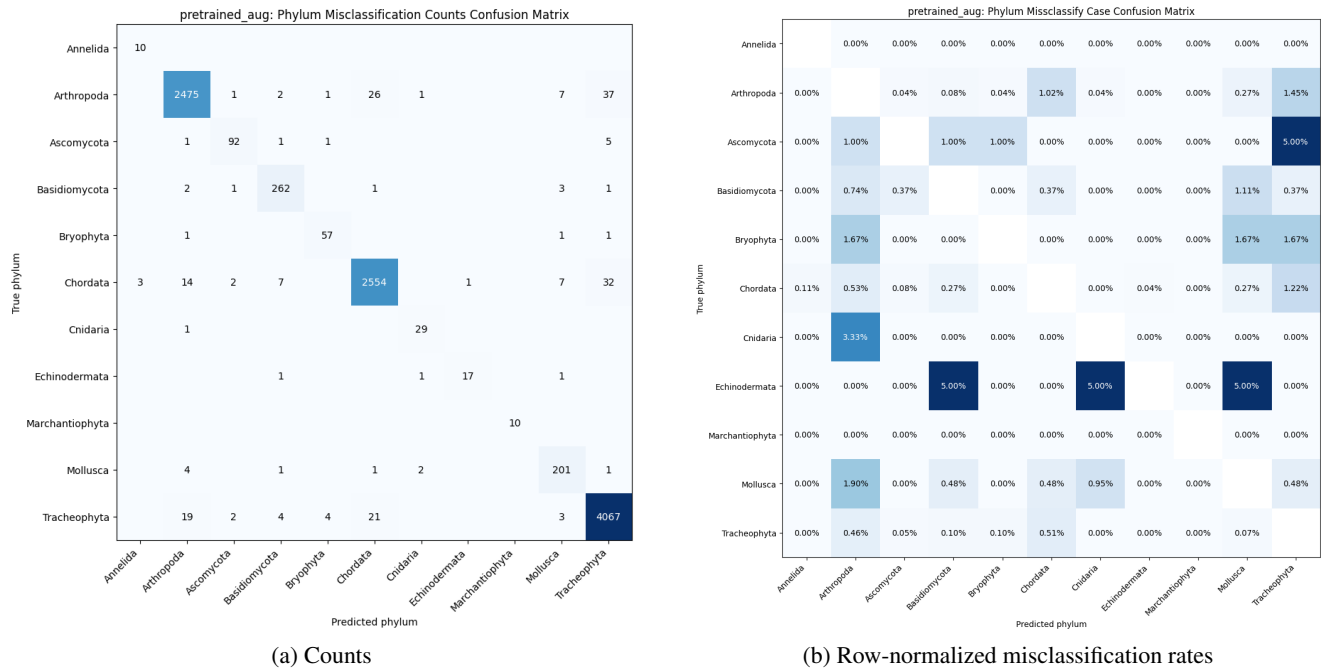
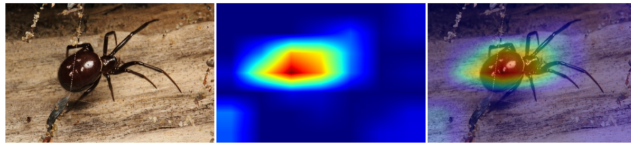


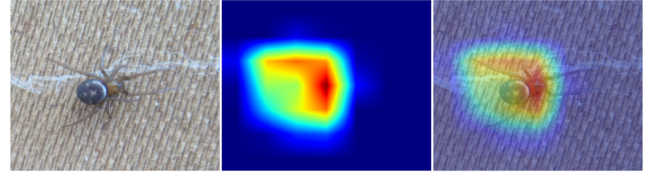
Figure 10. Phylum-level confusion matrices for ConvNeXt V2 experiment B. (a) Absolute counts including correct predictions on the diagonal. (b) Row-normalized off-diagonal misclassification rates.

15. Grad-CAM Confusable Species Pairs

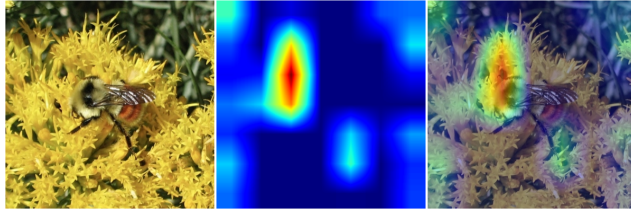
Fig. 10 illustrates Grad-CAM visual explanations across three confusable genus pairs from distinct biological classes: Spiders (*Steatoda*), Bumblebees (*Bombus*), and Mint Moths (*Pyrausta*), supplementing the analysis in Sec. 4.5.3.



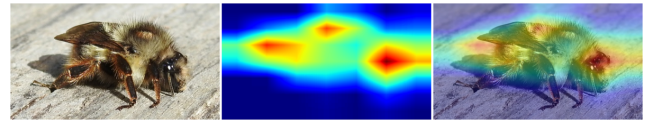
(a) *Steatoda capensis* (True: *S. capensis*, Pred: *S. capensis*, Prob: 0.3292)



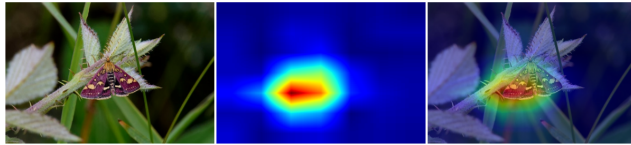
(b) *Steatoda grossa* (True: *S. grossa*, Pred: *S. grossa*, Prob: 0.3977)



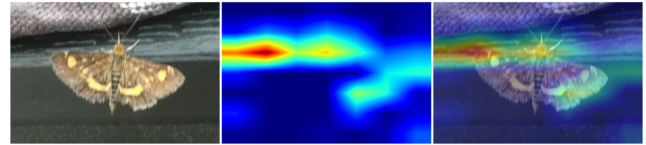
(c) *Bombus huntii* (True: *B. huntii*, Pred: *B. huntii*, Prob: 0.2010)



(d) *Bombus mixtus* (True: *B. mixtus*, Pred: *B. mixtus*, Prob: 0.8920)



(e) *Pyrausta purpuralis* (True: *P. purpuralis*, Pred: *P. purpuralis*, Prob: 0.3538)



(f) *Pyrausta aurata* (True: *P. aurata*, Pred: *P. aurata*, Prob: 0.4671)

Figure 11. **Grad-CAM comparison across three confusable genus pairs.** Top row: Genus *Steatoda* (Cobweb Spiders). Middle row: Genus *Bombus* (Bumblebees). Bottom row: Genus *Pyrausta* (Mint Moths).